

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 4 MODULE, AUTUMN SEMESTER 2013-2014

COMPUTER VISION

Time allowed TWO Hours

Candidates must NOT start writing their answers until told to do so

Answer THREE out of FOUR questions

No calculators are permitted in this examination.

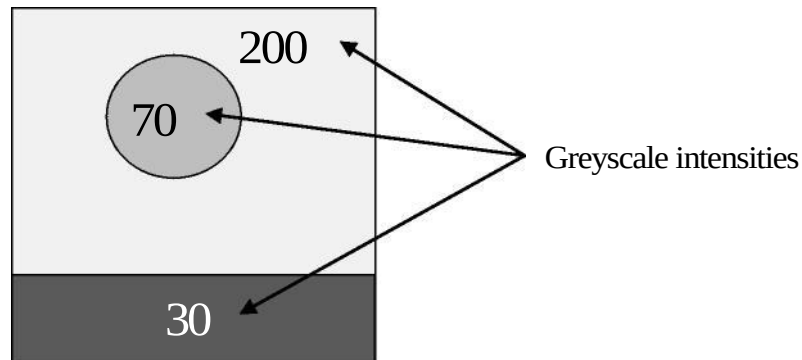
Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Answer all of the following sub-questions pertaining to Segmentation (25 marks):

- (a) Consider the image below. If one were to segment the image using k-means clustering with $k=2$ what would the segmentation look like? If on the other hand one were to use $k=3$ what would the segmentation look like? Draw a simple diagram for each segmentation case. Use these two cases to discuss some of the weaknesses of k-means clustering for segmentation. Discuss a simple heuristic that may be used in this case, to select the appropriate k .



(7 marks]

- (b) Consider the following 4x4 map of intensity gradients. Apply the watershed algorithm to this map of gradients, clearly labeling each step of the process. As a final result you should show a segmentation with one or more labeled regions.

(7 marks]

1	3	1	3
1	3	1	1
2	2	2	1
1	1	1	1

- (c) Specifically in the context of recognizing printed characters in scanned documents is the problem of segmentation ill-posed? Justify your answer.

(3 marks]

- (d) Consider a computer vision based system for analyzing a video stream for an auto-piloted car. Explain how you might use segmentation as an integral component of such a computer vision system.

(4 marks]

- (e) Briefly explain the graph-cuts algorithm for segmentation. Explain why cuts need to be normalized.

(4 marks]

2. Answer all of the following sub-questions pertaining to Motion Analysis (25 marks):

- (a) Consider the problem of counting vehicles on a highway using a simple frame-differencing approach and basic segmentation. Consider a very naïve frame-differencing approach where motion maps are obtained by subtracting the current and previous frames and normalizing. Explain three critical issues faced by this approach.

(5 marks]

- (b) Consider a computer vision system for digitizing a game of snooker for automated game analysis. The computer vision system is basically required to track snooker balls. Consider two high-level approaches to the problem: 1. segmentation followed by motion analysis and 2. motion analysis followed by segmentation. Firstly, discuss in detail how you would implement each approach. Secondly, discuss the pros and cons of each approach.

(6 marks]

- (c) Consider the problem of motion analysis when both the camera and the objects in the scene are moving. Based on the different techniques learnt in the lecture outline a possible computer vision solution to the problem of distinguishing camera motion from scene motion, which includes the computation of optical flow, and does not use any other sensors apart from one or more video cameras.

(6 marks]

- (d) Compare and contrast the concepts of “motion field” and “optical flow”.

(4 marks]

- (e) What is the aperture problem and why is it relevant to the computation of optical flow? How can the aperture problem be addressed?

(4 marks]

3. Answer all of the following sub-questions pertaining mainly to Tracking (25 marks):

- (a) Explain how both fixed and adaptive templates can lose track of their target objects?

(4 marks]

- (b) Explain in detail how you might apply snakes (or active contours) to the problem of tracking vehicles using optical flow maps (the snakes should be applied to the optical flow maps rather than the underlying frames).

(7 marks]

- (c) Consider two hypothetical trackers A and B, whereby tracker A is characterized by a good/strong motion model and a bad/weak appearance model whereas tracker B is conversely characterized by a bad/weak motion model and a good/strong appearance model. How do you expect trackers A and B to perform in the context of tracking people in a crowd? Justify your answer.

(5 marks]

- (d) Motion noise, occlusion, camouflage and object multiplicity are all challenges for any tracker. Explain the meaning of each one with one relevant example for each issue.

(6 marks]

- (e) Draw a simple diagram of the operation of a Kalman Filter.

(3 marks]

4. Answer all of the following sub-questions pertaining primarily to particle filters, scene analysis and Hidden Markov Models (25 marks):

(a) Explain the general operation of particle-filter based algorithms for tracking.
(6 marks]

(b) Consider an "ATM scene", whereby people typically walk up to an ATM machine, queue for a certain amount of time, perform their ATM transactions and then walk off. How would you propose to use a Hidden Markov Model (HMM) for modeling this scene in general, and for detecting any abnormal behaviour in particular?
(7 marks]

(c) Assume that someone has applied a Hidden Markov Model to a particular problem, and that the approach has failed to solve the problem. Can you conclude that Hidden Markov Model's are unsuitable for the problem? Justify your answer.
(6 marks]

(d) What does parameter tuning mean in the context of Hidden Markov Models?
(3 marks]

(e) How can Hidden Markov Models be used for recognizing different types of complex and motion-based behaviours?
(3 marks]